

SVALBARD, Svalbard and Jan Mayen

WMO: 010080

Lat: 78.246N

Lon: 15.466E

Elev: 27

StdP: 101.00

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 3	(b) -28.5	(c) -25.8	(d) -32.1	(e) 0.2	(f) -28.4	(g) -29.2	(h) 0.3	(i) -25.0	(j) 17.2	(k) -7.7	(l) 15.3	(m) -7.7	(n) 2.8	(o) 120	(p) 0.975

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 3.3	(c) 11.9	(d) 8.3	(e) 10.5	(f) 7.5	(g) 9.4	(h) 6.9	(i) 8.7	(j) 11.1	(k) 7.8	(l) 9.8	(m) 7.1	(n) 8.9	(o) 5.1	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.3	9.3	6.2	5.9	8.5	5.7	5.7	8.1	26.4	11.2	24.5	10.0	23.1	9.1	14.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.1	(b) 12.3	(c) 10.8	(e) -27.3	(f) 15.0	(g) 5.2	(h) 2.1	(i) -31.0	(j) 16.5	(k) -34.1	(l) 17.7	(m) -37.0	(n) 18.9	(o) -40.7	(p) 20.4		
			DB													
			WB	-27.6	10.7	5.1	1.4	-31.2	11.7	-34.2	12.6	-37.0	13.4	-40.7	14.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.7	-10.3	-11.1	-12.1	-8.9	-2.1	3.8	7.1	6.1	2.1	-3.5	-6.8	-9.1
	DBStd	8.52	7.83	7.50	7.00	5.87	3.65	2.21	2.02	2.19	3.05	4.64	5.64	6.46
	HDD10.0	5003	631	590	684	568	376	186	92	124	238	417	504	592
	HDD18.3	8039	889	823	942	818	634	436	347	381	488	676	754	851
	CDD10.0	5	0	0	0	0	0	0	3	2	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	5.0	5.9	5.7	5.3	4.5	4.3	4.3	4.6	4.0	4.2	5.1	5.6	6.0
(15) Precipitation	PrecAvg	203	17	18	20	11	7	9	18	24	22	17	19	20
	PrecMax	382	53	56	95	32	23	33	52	69	48	57	58	59
	PrecMin	84	4	1	4	1	2	1	1	4	4	3	3	2
	PrecStd	57	11	15	18	7	5	7	12	16	11	12	12	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.4	3.5	4.2	5.2	10.2	15.1	14.1	10.2	7.6	4.7	5.1
		MCWB	2.6	2.2	1.3	2.1	2.7	6.6	9.4	8.7	7.8	5.7	2.8	2.9
	2%	DB	2.9	2.8	1.8	2.4	4.0	8.8	12.5	11.3	8.4	5.7	3.4	2.7
		MCWB	1.2	1.0	0.0	0.7	1.9	5.8	8.5	8.2	6.3	3.9	1.5	1.0
	5%	DB	1.3	0.6	0.0	1.0	3.1	7.8	11.1	10.1	7.3	4.1	2.1	1.2
		MCWB	-0.3	-1.0	-1.3	-0.4	1.3	5.1	7.8	7.4	5.4	2.6	0.5	-0.4
	10%	DB	-0.2	-0.9	-2.1	-0.4	2.3	6.9	10.0	9.0	6.1	2.8	0.7	-0.2
		MCWB	-1.7	-2.3	-3.2	-1.6	0.6	4.4	7.2	6.8	4.5	1.5	-0.7	-1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.4	1.6	2.4	3.2	7.1	10.6	9.7	8.1	5.7	2.8	3.2
		MCDB	4.4	4.1	3.1	4.0	4.5	9.1	13.9	12.5	10.0	7.3	4.4	5.0
	2%	WB	1.3	0.9	0.0	0.9	2.2	6.1	9.1	8.5	6.5	4.0	1.6	1.1
		MCDB	2.8	2.6	1.6	2.2	3.6	8.2	11.6	10.8	8.1	5.4	3.1	2.5
	5%	WB	-0.2	-0.9	-1.4	-0.3	1.5	5.3	8.1	7.7	5.5	2.7	0.5	-0.2
		MCDB	1.2	0.6	-0.1	0.9	2.9	7.4	10.3	9.5	6.9	3.9	1.9	1.1
	10%	WB	-1.8	-2.3	-3.3	-1.7	0.6	4.6	7.4	7.0	4.7	1.5	-0.7	-1.5
		MCDB	-0.4	-1.0	-2.2	-0.5	2.2	6.5	9.4	8.5	6.0	2.6	0.6	-0.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.6	5.6	5.6	3.3	2.8	3.3	3.2	3.4	4.0	4.8	5.3
		MCDBR	5.8	6.0	6.7	5.4	3.5	4.1	4.9	4.7	4.6	4.6	5.3	6.2
		MCWBR	5.0	5.2	5.9	4.7	2.7	2.7	3.0	2.8	3.5	3.8	4.5	5.4
	5% WB	MCDBR	5.8	6.2	6.7	5.2	3.4	3.9	4.5	4.0	4.4	4.5	5.2	6.4
		MCWBR	5.0	5.5	6.0	4.7	2.8	2.7	2.9	2.6	3.6	3.8	4.6	5.7
(40) Clear-Sky Solar Irradiance	taub	N/A	0.187	0.244	0.278	0.298	0.292	0.297	0.286	0.237	N/A	N/A	N/A	N/A
	taud	N/A	1.899	2.061	2.192	2.214	2.323	2.424	2.425	2.329	N/A	N/A	N/A	N/A
	Ebn at Noon	0	32	561	771	825	857	835	776	627	0	0	0	0
	Edh at Noon	0	4	50	81	97	92	79	65	39	0	0	0	0
(44) All-Sky Solar Radiation	RadAvg	0.00	0.02	0.78	2.91	4.91	5.82	4.77	2.99	1.12	0.00	0.00	0.00	0.00
	RadStd	0.00	0.00	0.07	0.34	0.40	0.36	0.33	0.23	0.10	0.00	0.00	0.00	0.00

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+1.52	+4.63	+4.10	N/A	N/A	N/A	-552	-549	N/A	N/A	
(47) Regional (5 neighbors)		+1.52	+4.63	+4.10	N/A	N/A	N/A	-552	-549	N/A	N/A	

Nomenclature: See separate page